Part II: Analysis in Euclidean Spaces 08.14

Friday, August 14, 2026 8:53 PM

- Before we start:
  - Lecture recording of 08.13: Video is still converting.
  - No office hour this afternoon.

Recall from Yesterday:

$\lim_{k\to\infty} x_k = a \in \mathbb{R}^n$ equivalently $x_k \longrightarrow a$ $(k\to\infty)$ if:

For every $\varepsilon > 0$, there exists $N = N(\varepsilon) \in \mathbb{N}$. Such that

For all $k > N$: $\|x_k - a\| < \varepsilon$

- Some consequences of this defn:
  - When limit exists, it's unique:

Proof: If $x_k \longrightarrow a$ and $x_k \longrightarrow b$ as $k\to\infty$

$$\|a-b\| = \|(x_k-b)-(x_k-a)\| \le \|x_k-b\| + \|x_k-a\| \longrightarrow 0$$

as $k\to\infty$

This implies

$$\|a-b\| = 0 \iff a-b=0 \text{ so } a=b.$$

- Convergent sequence is bounded:

Proof. Let $x_k \longrightarrow a$ as $k\to\infty$. Pick $\varepsilon = 1$

Then by defn. there exists $N \in \mathbb{N}$ such that

$\|x_k - a\| < 1$ for all $k > N$

By $\|c\| - \|d\| \le \|c-d\|$ (Check this.)

$\|x_k\| - \|a\| \le \|x_k - a\| < 1$ for all $k > N$

This means: $\|x_k\| < \|a\| + 1$ for all $k > N$

This means: $\|x_k\| < \underline{\|a\| + 1}$ for all $k > N$

Take $M = \max\{\|x_1\|, \|x_2\|, \dots, \|x_N\|, \|a\| + 1\}$

Then $\|x_k\| \leq M$ for all k.

- $x_k \longrightarrow \infty$ as $k \to \infty$ if:

  For all $M > 0$, there exists $N = N(M) \in \mathbb{N}$ such that:

  For all $k > N$: $\|x_k\| > M$.

- A subsequence of $\{x_k\}$:

  $\{x_{k_j}\}_{j=1}^{\infty}$ where $k_1 < k_2 < k_3 \dots$

Sequence: $x_1, x_2, x_3, x_4 \dots x_k, x_{k+1}, x_{k+2} \dots$

Subsequence: x_{k_1} $x_{k_2} = x_{k_3}$ $\dots$ x_{k_j} $x_{k_{j+1}}$

- What does it mean formally that $x_k \not\to a$ $(k \to \infty)$.

  There exists $\varepsilon_0 > 0$, for every $N \in \mathbb{N}$

  There exists $k_0 > N$ such that

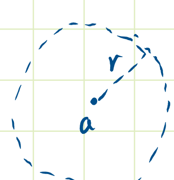
  $\|x_{k_0} - a\| \geq \varepsilon_0$.

- Remark: Make yourself very comfortable with the formal definition of convergence!!

- Overview of Point-Set Topology in $\mathbb{R}^n$.

  - Open Ball:

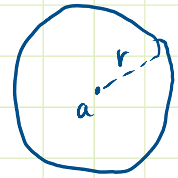
    $B_r(a) = \{x \in \mathbb{R}^n : \|x - a\| < r\}$



$B_r(a)$.

Closed Ball:

$$\overline{B_r(a)} = \{x \in \mathbb{R}^n : \|x-a\| \leq r\}$$

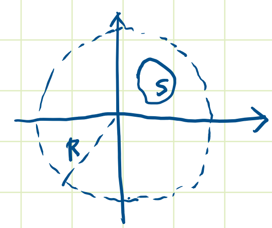


$\overline{B_r(a)}$

$$B_r(a) \subsetneqq \overline{B_r(a)}$$

- A set $S \subseteq \mathbb{R}^n$ is bounded if there exists $R>0$ such that

$$S \subseteq B_R(0).$$

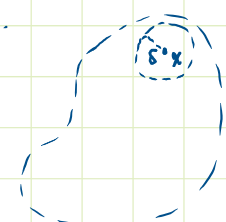


$$\Updownarrow$$

There exists M: $\|x\| \leq M \quad \forall x \in S$

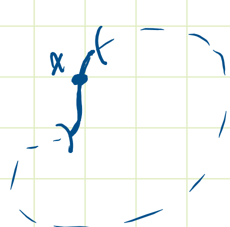
- Open Set: A set $U \subseteq \mathbb{R}^n$ open if for every $x \in U$ there exists $\delta > 0$:

$$B_\delta(x) \subseteq U.$$



Closed Set: A set $F \subseteq \mathbb{R}^n$ is closed if $F^c = \mathbb{R}^n \setminus F$ is open.

Claim: A potato with skin in $\mathbb{R}^n$ is not open.



Not open

Not closed

- There are sets that are open and closed. $\mathbb{R}^n$.

"Clopen" in topology.

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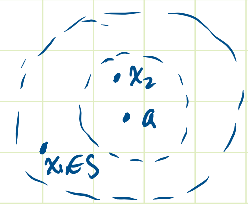
By convention: $\mathbb{R}^n$ and $\emptyset$ are clopens.

- Limit Points / Accumulation Points:

  a is a limit point of S if for every $\delta > 0$.

  $$B_\delta(a) \setminus \{a\} \cap S \neq \emptyset.$$



Consequence: a is a limit point

$\Updownarrow$

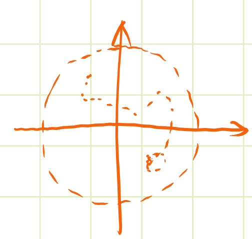
There exists $\{x_k\}_{k=1}^{\infty} \subseteq S : x_k \to a \ (k \to \infty)$

- Compactness:

  Fact: (Bolzano - Weierstrass Theorem)

  Every bounded sequence in $\mathbb{R}^n$ has convergent subsequence.

  In $\mathbb{R}$: $a_n = (-1)^n$.



- Compact Set:

  Let $K \subseteq \mathbb{R}^n$

  A collection of open sets $\{U_\alpha\}_{\alpha \in A}$ is an open cover of K if

A collection of open sets $\{U_\alpha\}_{\alpha \in A}$

open cover of K if
$$K \subseteq \bigcup_{\alpha \in A} U_\alpha$$

Note: The index set A might be uncountable.

The set K is compact if every open cover of K contains a finite subcover:

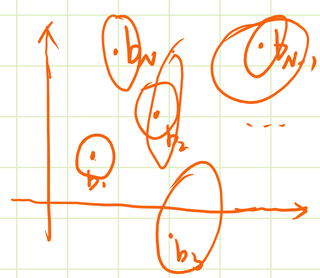
Whenever $K \subseteq \bigcup_{\alpha \in A} U_\alpha$ there exists

$\alpha_1, \alpha_2, \dots \alpha_m$ m finite, such that

$$K \subseteq \bigcup_{i=1}^{m} U_{\alpha_m}.$$

- Why do we need compactness?

Finite set: $B = \{b_1, b_2, \dots, b_N\} \subseteq \mathbb{R}^n$



Open cover

$\downarrow$

Finite subcover

Compactness is "fintely infinite". for arbitrary sets.

- Heine-Borel Theorem:

$K \subseteq \mathbb{R}^n$ compact $\Leftrightarrow$ Closed and bounded.

$\mathbb{R}$ is NOT compact: $\{(-n, n)\}_{n=1}^{\infty}$ open cover for $\mathbb{R}$

NO Finite Subcover.

- $\mathbb{R}$ is NOT compact: $\{(\ldots)\}_{n=1}$ ...

NO Finite Subcover.

$(0,1) \subseteq \mathbb{R}$ is NOT compact: $\left\{\left(0, 1-\frac{1}{n+1}\right)\right\}_{n=1}^{\infty}$ open cover for $(0,1)$

No finite Subcover.

- In $\mathbb{R}^n$, TFAE:

(1). $K \subseteq \mathbb{R}^n$ is compact.

(2). (Sequential Compactness):

Every sequence $\{x_k\} \subseteq K$ has a convergent subsequence whose limit belongs to K.

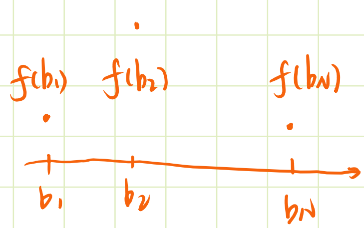
(3). (Heine-Borel):

K is closed and bounded.

Fact: Closed sets contain their limit points

(Prove by definition and look at the complement)

- $B = \{b_1, b_2, \ldots, b_N\}$ Finite $f: B \to \mathbb{R}$



$\max_{b \in B} f(b)$ ✓ : Rank $f(b_1) \cdots f(b_N)$.

Since $\{f(b_i)\}_{i=1}^{N}$ finite.

K compact $\Rightarrow$ $\max_{x \in K} f(x)$ ✓

Useful for establishing the existence of a maximum

$x \in \mathbb{R}$

Useful for establishing the existence of a maximum

- Limits and Continuity of Mappings:

$$f: D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m.$$

Let a be a limit point of D and $b \in \mathbb{R}^m$

$f(x) \longrightarrow b$ as $x \to a$

if for every $\varepsilon > 0$, there exists $\delta = \delta(\varepsilon) > 0$ such that

for all $x \in D$, $0 < \|x - a\| < \delta$ ($\Leftrightarrow x \in B_\delta(a)$)

We have $\|f(x) - b\| < \varepsilon$

This is the ε-δ Language for $\lim\limits_{x \to a} f(x) = b$

We can also characterize the limit using sequence

Heine Principle: Let a be a limit point of D

Then $\lim\limits_{x \to a} f(x) = b$

$\Updownarrow$

For every sequence $\{x_k\} \subseteq D \setminus \{a\}$ such that

$x_k \to a \quad (k \to \infty)$.

we have $f(x_k) \longrightarrow b$

- Why is this useful?

Example: Show $f(x,y) = \dfrac{xy}{x^2 + y^2} \quad (x,y) \neq (0,0)$

$\lim f(x,y)$ doesn't exist.

$$\lim_{(x,y)\to(0,0)} f(x,y) \text{ doesn't exist.}$$

Along the line $y = kx$ we have:

$$f(x,y) = f(x,kx) = \frac{k}{1+k^2}$$

Different values of k will give a different limit. So the limit doesn't exist.

Checking lines like $y=kx$ is NOT always enough.

Example': $g(x,y) = \dfrac{x^2 y}{x^4+y^2}$ $\quad (x,y) \neq (0,0)$

Along any straight line $(x,y) = (\alpha t, \beta t)$

$$g(\alpha t, \beta t) \to 0 \quad \text{as} \quad t \to 0 \Leftrightarrow (x,y) \to (0,0)$$

But if we take $y = x^2$

we have $g(x,x^2) = \frac{1}{2} \neq 0$

So the limit doesn't exist.

- Limits preserves arithmetics when defined.

  $+, -, \times, \div$ by non-zero. when $f: D \to \mathbb{R}$

- Continuity of Mappings:

  Let $f: D \subseteq \mathbb{R}^n \to \mathbb{R}^m$, $a \in D$.

  f is continuous at a if for every $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that there exists $\delta = \delta(\varepsilon) > 0$ such that

for every $x \in D$, $\|x-a\| < \delta$ we have

$$\|f(x) - f(a)\| < \varepsilon$$

If a is a limit point of D. then by the definition of limits, this is equivalent to

$$\lim_{x\to a} f(x) = f(a) = f\left(\lim_{x\to a} x\right).$$

Note: By convention, if a is isolated, then f is continuous at a.

If f is continuous at every point $a \in D$, we say f is continuous on D.

If: $f: D \to \mathbb{R}$. arithmetics preserves continuity.

- Local Sign Preservation:

Let $f: D \to \mathbb{R}$ continuous at $a \in D$. If $f(a) > 0$, then there exists $\delta > 0$ such that

for all $x \in D$, $\|x-a\| < \delta \Rightarrow f(x) > 0$.

Proof: Take $\varepsilon = \frac{1}{2} f(a)$ by continuity, $\exists \delta > 0$:

$$\|x-a\| < \delta \Rightarrow |f(x) - f(a)| < \frac{f(a)}{2}$$

This implies

$$\underline{-\frac{1}{2} f(a) < f(x) - f(a)} < \frac{f(a)}{2}$$

$$\Rightarrow \quad f(x) > \frac{1}{2} f(a) > 0.$$

$\square$

- Note: If a is a limit point of D then by the Heine Principle:

$$x_k \to a \quad \text{then} \quad f(x_k) \to f(a), \quad k \to \infty.$$

- Composition of continuous maps are continuous:

$$D \subseteq \mathbb{R}^n \xrightarrow[\substack{\text{cont.} \\ \text{at } a \in D}]{g} U \subseteq \mathbb{R}^m \xrightarrow[\substack{\text{cont.} \\ \text{at } g(a) \in U}]{f} \mathbb{R}^p \qquad f \circ g : D \subseteq \mathbb{R}^n \to \mathbb{R}^p$$

then $f \circ g$ is continuous at a

Proof: Let $x_k \to a$. then $g(x_k) \to g(a)$ by cont. of g

then $f(g(x_k)) \to f(g(a))$ by cont. of f.

$\square$

- Topological Characterization of continuity.

  - Preimage: $f : D \to \mathbb{R}^m$. Let $A \subseteq \mathbb{R}^m$

The preimage of A under f is:

$$f^{-1}(A) = \{x \in D : f(x) \in A\}$$

If $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$, take $A = \{0_{\mathbb{R}^m}\}$

$$\ker T = \underbrace{T^{-1}(A)} = \{x \in \mathbb{R}^n : T(x) = 0_{\mathbb{R}^m}\}$$

This does. This doesn't show up in linear algebra.

This does. This doesn't show up in linear algebra.

- Fact: (Check for yourself):

$$f^{-1}(A^c) = [f^{-1}(A)]^c$$ Hint: Show $\subseteq$ and $\supseteq$

- Let $f: D \subseteq \mathbb{R}^n \longrightarrow \mathbb{R}^m$. TFAE:

(1). f is continuous on D.

(2). for every open set $U \subseteq \mathbb{R}^m$ the preimage $f^{-1}(U)$ is open in $\mathbb{R}^n$.

(3). for every closed set $F \subseteq \mathbb{R}^m$ the preimage $f^{-1}(F)$ is closed in $\mathbb{R}^n$.

"Continuity means that preimage of open set is open".

Proof: (2) $\Leftrightarrow$ (3) by defn. of open/closed sets and fact.

(1) $\Rightarrow$ (2): Take $a \in f^{-1}(U)$. by defn. of preimage.

$$f(a) \in U.$$

U is open. there exists $\varepsilon > 0$:

$$B_\varepsilon(f(a)) \subseteq U$$

By continuity of f: there exists $\delta = \delta(\varepsilon) > 0$ for all $x \in D$, $\|x - a\| < \delta$ we have

$$f(x) \in B_\varepsilon(f(a)) \subseteq U.$$

This means $\forall x \in B_\delta(a)$ $f(x) \in U$

This means $\forall x \in B_\delta(a)$, $f(x) \in U$

Thus $B_\delta(a) \subseteq f^{-1}(U)$. therefore

$f^{-1}(U)$ is open.

(2) $\Rightarrow$ (1): Suppose now preimage of open set is open.

Fix $a \in D$ and $\varepsilon > 0$.

$B_\varepsilon(f(a))$ is open, therefore

$f^{-1}(B_\varepsilon(f(a)))$ is open and contains a.

By openness, there exists $\delta > 0$, such that

$B_\delta(a) \subseteq f^{-1}(B_\varepsilon(f(a)))$.

Hence for all $x \in D$, $\|x - a\| < \delta$

We have $\|f(x) - f(a)\| < \varepsilon$.

This gives us continuity. $\square$.

Coming Next in Person:

- Continuous functions on a compact set.
- Uniform Continuity
- Differentiation.